

House Price Prediction Using Linier Regression

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Abstract— Property entrepreneurs will compete to build properties, especially houses for investment facilities. This will make house prices increase day by day with a high purchasing power of the people. Consumers will think in buying a house whether the house they buy will have a good profit value or not. The aim of the study is to build a model that may predict house price for company and become a business decision for consumers. The methodology used seven major steps namely, business understanding, data understanding, data cleaning, data standardization, modelling, and evaluation. This study develop a linear regression model for home price prediction and tests it using data from the Maribelajar company. These are carried out on the Azure Platform by creating two pipelines. One pipeline for training and another for testing. The results are then visualized using Power Business Intelligence for providing a proper business performance analysis. From the experimental results, the model achieved RMSE= 0.0334 and R coefficient = 0.7. The data analysis and testing in this study show that the multiple linear regression model can forecast and evaluate housing prices to some extent, but the algorithm may still be improved using more advanced machine learning approaches.

Keywords—Azure Machine Learning, House Price Prediction, Linear Regression, Machine Learning, Predictive Analysis

I. INTRODUCTION

Prediction is a key task in science and engineering. Scientists utilize various methods to discover patterns in a given dataset to predict future-related parameters. One of the most important tasks in machine learning is prediction [1]. The common predictive approach used by researchers is regression. It is frequently applied to house price prediction. When property company want to promote their products and attract customers for buying house, this doesn't happen automatically because of the complexity of the customer. Experts predict that housing affordability will continue to decline and will limit opportunities to own a home in the future [2]. Property entrepreneurs will compete to build properties, especially houses for investment facilities. This will make house prices increase day by day with a high purchasing power of the people. Consumers will think in buying a house whether the house they buy will have a good profit value or not. In investing, it cannot be separated from guessing the ups and downs of prices with the aim of not losing money. Uncertain and unpredictable price makes investors or home buyers need a system to predict house prices based on the location of the house [3].

However, house price marketing has some drawbacks. For example, when you reach out to customers who are confused about choosing a home, it's a waste of time, and they find it annoying because it doesn't match their needs and wants. In addition, it can trigger a negative attitude towards the company if the customer feels aggrieved. House price is

influenced by several factors such as the number of attributes, characteristics, and environment. The most usual variables for the house prices prediction model involve attributes such as type, the value of the property, and the number of bedrooms [3].

Machine Learning is widely used by creating a training model for computers to make predictions, draw conclusions, refine, analyze, develop, and test models [4]. The hallmark of machine learning is that there is a training and learning process known as data training and data testing to test the data model. One of the most common analytical techniques used in machine learning is predictive analysis. It is carried out on data to extract information and predict trends and behaviour patterns [5]. Predictive analysis is likely to determine the outcome of an event and in the future or even the likelihood of conditions that will occur and the trends that can affect it. It uses statistical knowledge to forecast future events based on the assumption that what has occurred in the past may have an influence on future events [6]. The goal is to generate relevant information, reliable explanation, better outcomes, more intelligent decisions, and predict future events by analyzing the speed, volume, and value of big data that will enforce each organization in transforming data into accessible information, evaluate the risk before it occurs, and recommend the best planning for the future [7].

Targets potential property company clients who are interested in buying a home and will reduce marketing costs by saving wasted campaign effort and resources. On the other hand, it is a relatively higher cost not to contact a potential client who is willing to invest than it is to contact a client who is not interested. Therefore, to maximize profits, maintain long-term relationships with consumers, and reduce marketing costs, companies are required to identify profitable customers. House price prediction has an impact on those who are likely to have a house that is by the consumer's financial condition and consumer needs [8]. The goal of this study is to predict house price using linear regression.

II. RELATED WORKS

Azimlu *et al.*, [1] stated house price prediction using clustering and genetic programming along with conducting a comparative study. This methodology showed that clustering algorithms are capable of not only predicting house prices but also enhancing the prediction accuracy and decreasing the computational cost at the same time. The different from this study is the using of linear regression algorithm.

Rao [9] stated stock prediction analysis by using linear regression machine learning algorithm. Linear regression is also executed on stock market values. It is trained using historic stock market values data to make predictions on future prices. The results may offer guidance for people on when to invest money in the stock market. The model was developed

and implemented in Python. This study used house price prediction as the difference.

Rath *et al.*, [10] stated terms of the health domain, the linear regression algorithm may assess the relative impact of new active cases of coronavirus disease and forecast the number of active cases in the next coming days. These models gain incredible accuracy in Covid-19 recognition. The different on this study used house price prediction.

Adetunji *et al.*, [11] stated house price prediction using random forest machine learning technique. The result is the model can be used to predict house prices, however several other machine learning models can also be explored for house price prediction. The different on this study used Linier Regression algorithm.

Aggarwal *et al.*, [12] stated an insight into machine learning techniques for predictive analysis and feature selection. This study can give a good heads up to whoever wants to know and use predictive analysis for the academic or business application. This study confirms linear regression as machine learning techniques that normally is used for forecasting. The different on this study used house price prediction.

Rahayuningtyas *et al.*, [13] stated house price prediction using general regression neural network. The results show that prediction using GRNN is able to produce a fairly good prediction accuracy. This study used Linear Regression as the difference.

III. METHODOLOGY

A. Dataset

We used the maribisnis dataset. The dataset comprises a year of house price data from 2014 to 2015. The variables are price, bedrooms, bathrooms, sqft_living, sqft_lot, floors, grade, sqft_above, sqft_basement, zip code, lat, long, and sqft_living15. It is obtained from the Maribelajar company.

B. The Proposed Methodology

The proposed methodology is displayed in Fig. 1.

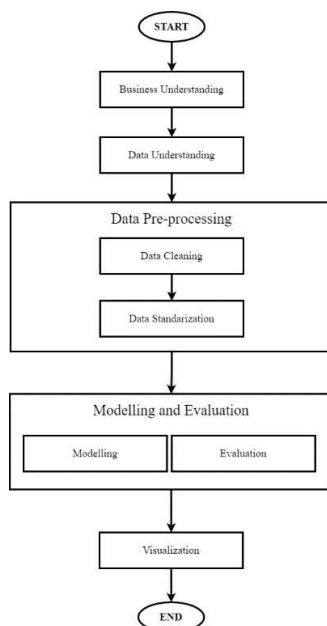


Fig. 1 The proposed methodology

1) *Business understanding*: This phase comprises a highly clear statement of the problem as well as techniques for assessing goal attainment. There are two cycles in this phase, namely determining business objectives and characterizing questions that can be addressed from the existing dataset. In this work, the business objective is how to forecast the price of the other houses based on the available price of certain houses.

2) *Data Understanding*: after understanding the business value, the next step should apply to the goal, therefore the researchers can examine it and guarantee that the information can be altered accurately. The dataset is in the CSV format, consists of 19,906 rows and 14 columns. These infer that Sqft_Living is the region inside the house, Sqft_Lot is the region outside the house, Grade is the general rating of the house in light of a rating framework, sqft_above is the region other than the basement, sqft_basement is the region of house basement, and sqft_living15 is the region inside the house in 2015.

3) *Data pre-processing*: It is utilized to turn information into a valuable and effective arrangement. The interaction includes approving and figuring information. The objective is to assess and fix the exactness of the incorrect information. An example of data processing is the interaction of removing segments and lines, changing dates on a dataset, etc.

4) *Data Cleaning*. In this cycle, various activities are conducted, including filling in missing values and eliminating unnecessary features. Eliminating sections containing outrageous qualities that can affect the accuracy of information demonstrations.

5) *Data Standardization*: The module is used to convert the output to CSV. Assemble information mining work processes to observe the selection of the calculation boundary's setting and conduct mining operations on preprocessed data.

6) *Modelling*: Machine Learning has been employed in many sectors in past decades like image processing, pattern recognition, medical diagnosis, predictive analysis, and product recommendation. Machine Learning models are built using Linear Regression on the Machine Learning Designer. Linear regression is a statistical procedure for calculating the value of a dependent variable from an independent. Machine Learning is carried out by making a workspace in the Azure portal where you can create a machine-learning studio, create process cases and bunches, transfer the required datasets, and create pipelines based on Fig. 2 (training pipeline) and Fig. 3 (testing pipeline).

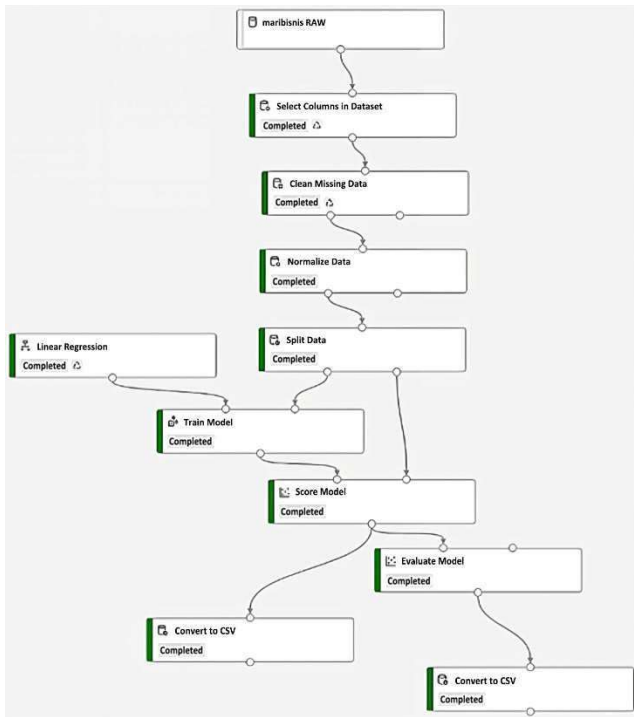


Fig. 2 Model training pipeline using linear regression
The explanation of each module in Fig. 2 is explained in Table 1.

TABLE I. LIST OF TRAINING PIPELINE MODULES

Modules	Description	Configuration
Select Columns in a dataset	Select columns that include or exclude from a dataset in an operation	Select Columns: All Columns except Date
Clean Missing Data	A module for dealing with missing values in datasets. This tool is optional, but it is best to place this tool.	Column Names: All Columns except Date
Normalizing data	A module that restricts dataset values to a standard range	Column Names: All Column except date
Split data	A module that divides the rows of a dataset into two distinct sets	Fraction of rows: 0.8 for training and 0.2 for testing Random Seed:123
Train Model	A module to train a classification or regression model in a supervised manner	Select Column: Price
Score Model	A module for scoring predictive regressions	
Evaluate model	A module for evaluating the results of regression models using standard metrics	
Convert to CSV	A module that converts data input to a comma-separated value format	

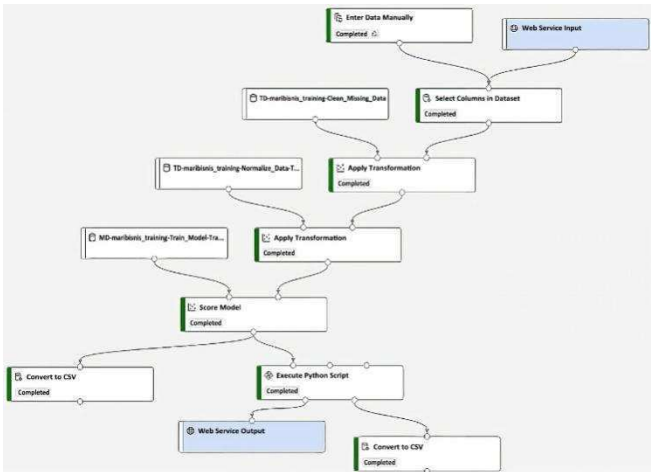


Fig. 3 Model testing pipeline
Details of modules in Fig. 3 can be seen in Table 2.

TABLE II. LIST OF TESTING PIPELINE MODULES

Modules	Description	Configuration
Enter Data Manually	A module for testing the data manually	Syntax Code: price, bedrooms, bathrooms, sqft_living, sqft_lot, floors, grade, sqft_above, sqft_basement, zipcode, lat, long, sqft_living15 221900, 3, 1, 1180, 5650, 1, 7,1180, 0,98178, 47.5112, -122,257, 1340
Select Columns in the dataset	Select columns that include or exclude from a dataset in an operation	Select Columns: All Columns except Date
Score Model	A module for scoring predictive regressions	-
Execute Python Script	A module for evaluating the results of regression models using standard metrics	-
Convert to CSV	A module that converts data input to a comma-separated value format	-

7) *Evaluate*: In this process, MAE is used according to some sources. MAE is the variance of the model's predicted value to the actual value. The smaller the variance, the better the model's prediction. RMSE is measured in the same unit as the label with lower values indicate a better model. RSE is a relative metric between 0 and 1 that is determined by the square of the difference between actual and predicted value. Based on the absolute differences between predicted and actual values, RAE is a relative measure between 0 and 1. The closer RAE is to 0, the more accurate the model. R coefficient is the percentage of variance explained by the model between predicted and actual values. Thus, the closer this value is to 1, the better this model performs.

C. Power Business Intelligence

Power Business Intelligence is a suite of business intelligence and analytics tool for analyze data, share insights and gets answers quickly with the help of interactive data visualization using dashboard available on every device such

as Applications, Desktops, Mobile, etc [14]. Power Business Intelligence is applied to visualize results for giving a proper business analysis performance. Three types of graphs are selected namely, card price, line cart, and map. Card price presents the predicted value of the house price. The most intuitive display, line chart, is for house price forecast while the map is used to understand the situation around the target market.

IV. RESULTS & DISCUSSION

The result of business understanding produces a specific way of thinking, a big perspective, and appropriate design thinking, such as defining business goals, developing business models, and selecting target businesses. Business Understanding aims to use Azure Machine Learning to predict house prices. The final number of columns used for modelling is 13 columns.

A preprocessing step provides the dataset with some processing steps that will ensure no missing values, no errors, and the dataset will be processed more efficiently. It can develop further precision by modifying datasets, orchestrating segments and lines, and so on. It is necessary to display the data to utilize the model of direct relapse for predicting house costs. A successful interaction requires that no errors are made and that a scoring model is obtained.

According to the results from modelling and evaluation data, the expected house price values are around \$382.46, RMSE 0.033678892, MAE 0.023183424, RSE 0.3, RAE 0.519715453, and R Coefficient 0.696 = 0.7. Comparison of research using previous research conducted by Rahayuningtyas *et al.*, with the title “Prediksi Harga Rumah Menggunakan General Regression Neural Network” [13]. The resulting RMSE score is 7.66. The result from this study of the Linear Regression accuracy test with an RMSE of 0.0334, the Linear Regression is better than the model used in the comparative study.

This study further confirms the power of linear regression machine learning technique in predicting the prices of a house based on variables made available in Maribelajar Company dataset. It is reaffirmed by adetunji *et al.*, [11] stated this will assist potential buyers in determining the right time to buy a home. The resulting RMSE score is 2.58.

This study also provide clearer interpretation of the findings, the house trend, general predicted house price, and daily predicted house price are presented in Fig. 4, Fig. 5, and Fig. 6.

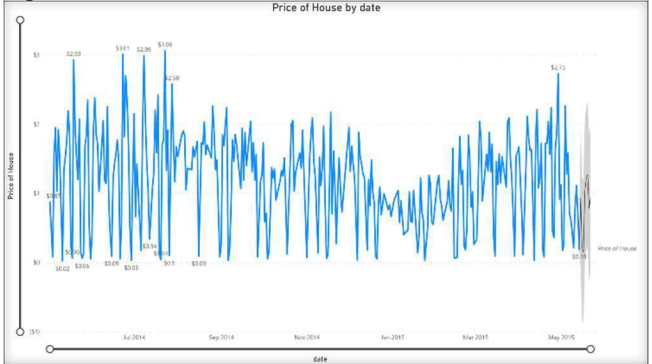


Fig. 4. House price prediction by date

Fig. 4 shows the resulting graph representing the development of house prices throughout 2014 – 2015 where

prices often decline but according to forecasts the gray shaded area will increase the next day. This shows that house prices can be predicted well.

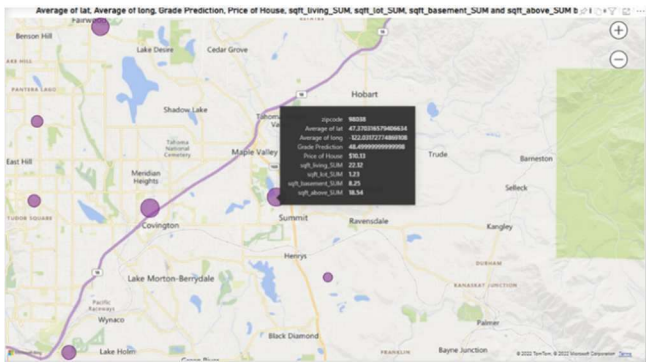


Fig. 5. House Trends

Fig. 5 shows a map visualization represents house trend mapping trends with its characteristics such as grade, price, outside and inside area, basement area, and area besides basement. that a house has a passing grade, a reasonable price, an adequate area inside the house, a wide enough outside of the house, and others so it can be an agreeable house.



Fig. 6. House price prediction

Fig. 6 displays a card price visualization which represents the house price prediction with good performance modelling. shows the number of house price predictions, which amounts to an average house price prediction of \$382.46, which can affect organizations to sell a house.

This study confirms the simplicity for showing the results It agrees with Bhargava *et al.*, [14] stated visualization of data is a very important thing in every organization and Business development, by means of visualization the end user or user can read the data easier and convenient to understand the data.

V. CONCLUSION & RECOMMENDATION

In this study, different prediction model schema use Linear Regression. The house price prediction based on Linear Regression is \$382.46. The R coefficient indicates that the model has the accuracy and performance required for optimal prediction. In this case, the most suitable values are RMSE 0.033678892, MAE 0.023183424, RSE 0.3, RAE 0.519715453, and R Coefficient is 0.696 = 0.7. According to the visualization of the trend, the house has a large sqft_living area, a high-quality house can be seen from the Grade, and the condition of the area can be seen from the location mapping. Recommendations for future study are different algorithms that matched the time-series data will be used to get a higher R Coefficient and the ability to use more data, work on larger data sizes (sales, payments, etc.) may be considered for addition, and having more numeric attributes can bring improvement in prediction.

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